

integer field, whilst the value could be a number (say an age), a string (say a name), or a picture (say a photo). All these disparate data could be bundled into a single table and be processed.

In addition to being flexible and simple, these databases can also scale extremely well.

Open source has been most dominant in these databases.

Two interesting DBs in this space are Voldemart and Riak. Voldemart, named after the Harry Potter villain, is one of the most deployed KV stores and is used extensively by LinkedIn. Voldemart brings in parts of ACID to the KV stores. It has a good storage layer and also supports MVCC as a consistency model. This makes Voldemart a complete database. Riak is another interesting database in the KV store space because of the way it has adopted the Web language as its front-end in lieu of SQL.

Document stores: Another evolving area in the non-relational database space is 'document stores'. These stores, as the name suggests, store documents similar to the way an RDBMS stores rows in a table. It is essential to understand that the notion of a document is similar to a row in the RDBMS and not a table. Document stores can potentially store different kinds of document types like XML, Word documents, etc. Other than the ability to store, document stores also have the capability to query parts of the document as if they are column values of an RDBMS.

The most famous open source implementation of the document store is MongoDB, which brings certain RDBMS concepts to the document stores. This includes indexing and ad-hoc querying. This makes MongoDB very attractive for RDBMS migrations for document storing purposes.

The 'clustered' DB space

Along with the NoSQL explosion, the other critical trend to dominate database growth has been 'horizontal scalability'. A few years back, the focus was always on vertical scalability, but with Google's bigtable research and Amazon's dynamo research along with the use case of big-data, horizontal scalability is the in-thing. This requirement has also brought out the famous CAP theorem, which argues that C (Consistency), A (Availability) and P (Partitioning) cannot be achieved at the same time. One or more of them has to be sacrificed. This has led to different models—a famous one is called 'eventual consistency'. Horizontal scalability offered in the database space is through clustered databases. These concepts are mentioned in this article to interest the reader to go back and dig deeper.

Relational clustered space: The relational database space has adopted clustering seriously. One of the best adoptions of clustering in the relational space is by MySQL, in its NDB cluster implementation. This is a 'shared-nothing' implementation, which offers scalable clustering. Another interesting attempt at clustering is by a MySQL variant called Drizzle. The third one is not a full database but a replicator, known as the tungsten-replicator. This works on the MySQL database to provide very fast and scalable replication.

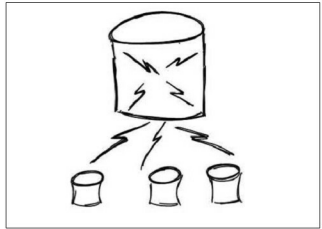


Figure 5: Database distribution

One of the most popular open source relational clustered databases is called VoltDB and falls in the category called NewSQL. OSFY carried a detailed article on NewSQL in its February 2012 edition.

Non-relational clustered space: Unlike the relational database space, the current non-relational database space always comes with horizontal scalability requirements. Hence, most of the NoSQL databases scale naturally. The most interesting of all in the clustered non-relational space is HBase. The intriguing part is the way Hadoop is used to produce this scalability. Cassandra, probably the most famous of all the NoSQL databases, is a columnar clustered scalable database.

Things to watch

A lot of technologies, jargon and open source products have been introduced here, to bring about reader awareness. The database space continues to expand rapidly. One of the areas to really watch out for is the OLTP+OLAP merger brought about by SAP HANA. Another area to watch is big-data analytics and what open source databases may want to offer in this space.

The complexity now is that there are too many database product options that have very marginal differences. The current state is such that most of the databases are building blocks. A DB like Riak offers only part of the database, and a store like Memcached needs to be used with it; sometimes, an external replicator also needs to be used. This is quite different from an integrated Big-Bang solution like, say, Oracle RAC. This integration makes these systems flexible, but it also makes the space a bit more technical. So I expect there will be binding products or natural plug-in architectures that may bring these databases together. 

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